

9-18

Matrix Representation of a Linear Map

Notation: let $\vec{x} \in V$ and $B_V = \{\vec{v}_1, \dots, \vec{v}_n\}$ be a basis. If $\vec{x} = \sum_{i=1}^n x_i \vec{v}_i$ we denote the components of $\vec{x}$ w.r.t. B_V as the vector

$$[\vec{x}]_{B_V} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

Note that if $\vec{x} \in \mathbb{R}^n$, then by default $\vec{x} = [\vec{x}]_{\{\vec{e}_i\}}$

Theorem 7.1 let V and W be n and m dimensional vector spaces over F . Given $T \in \mathcal{L}(V, W)$ and basis $B_V = \{\vec{v}_1, \dots, \vec{v}_n\}$ of V , there exists unique matrix $A \in F^{m \times n}$ such that

$$T(\vec{x}) = A \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}_{B_V}$$

for all $\vec{x} \in V$. This matrix is said to be the

"matrix representation of T w.r.t. $\mathcal{B}_V$ " and is denoted by $[T]_{\mathcal{B}_V}$.

Proof is by construction

Theorem 7.2 let $T \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ and let $\{\vec{v}_1, \dots, \vec{v}_n\}$ be a basis for $\mathbb{R}^n$. Then

$$[T]_{\{\vec{e}_i\}} = [T\vec{v}_1 \dots T\vec{v}_n] [\vec{v}_1 \dots \vec{v}_n]^{-1}$$

Proof: let $\vec{x} \in \mathbb{R}^n$ be such that $\vec{x} = \sum_{i=1}^n x_i \vec{v}_i$.

$$\text{Then } T(\vec{x}) = \sum_{i=1}^n x_i T(\vec{v}_i) = [T\vec{v}_1 \dots T\vec{v}_n] \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}.$$

$$\text{Furthermore as } \vec{x} = [\vec{v}_1 \dots \vec{v}_n] \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix},$$

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = [\vec{v}_1 \dots \vec{v}_n]^{-1} \vec{x} \text{ as desired.}$$

Corollary 7.3 If $T \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$ then

$$[T]_{\{\vec{e}_i\}} = [T\vec{e}_1 \dots T\vec{e}_n]$$

Corollary 7.4 If $T \in \mathcal{L}(\mathbb{P}_n, \mathbb{P}_m)$ and $\{e_i(x)\}$

denotes the standard basis where $e_i = x^i$, then

$$[T]_{\{e_i(x)\}} = \begin{bmatrix} (Te_0)_0 & \cdots & (Te_n)_0 \\ \vdots & & \vdots \\ (Te_0)_m & \cdots & (Te_n)_m \end{bmatrix}$$

where $(Te_i)_j$ is the coefficient of x^j in $T(x^i)$.

Example: As $\mathbb{P}_n \cong \mathbb{R}^{n+1}$ by $e_i(x) \rightarrow \vec{e}_{i+1}$

$$x^3 + 8x + 5 \longrightarrow \begin{bmatrix} 5 \\ 8 \\ 0 \\ 1 \end{bmatrix} \in \mathbb{R}^4$$

$5e_0 + 8e_1 + e_3$

Exercise:

- 1) Letting $T \in \mathcal{L}(\mathbb{P}_3, \mathbb{P}_3)$ be s.t. $T(f) = f' + 4f$
find $[T]_{\{e_i\}}$

Acting on the standard basis we obtain

$$T(e_0) = (1)' + 4 \cdot 1 = 4 = 4e_0$$

$$T(e_1) = (x)' + 4 \cdot x = 1 + 4x = e_0 + 4e_1$$

$$T(e_2) = (x^2)' + 4 \cdot x^2 = 2x + 4x^2 = 2e_1 + 4e_2$$

$$T(e_3) = 3x^2 + 4x^3 = 3e_2 + 4e_3$$

$T_{e_0} T_{e_1} T_{e_2} T_{e_3}$

and thus $[T]_{\{e_i\}} = \begin{bmatrix} 4 & 1 & 0 & 0 \\ 0 & 4 & 2 & 0 \\ 0 & 0 & 4 & 3 \\ 0 & 0 & 0 & 4 \end{bmatrix} \begin{matrix} e_0 \\ e_1 \\ e_2 \\ e_3 \end{matrix}$

Def 7.5 let $\mathcal{B} = \{\vec{v}_i\}$ and $\mathcal{B}' = \{\vec{w}_i\}$ be two bases of V . We define the change of basis matrix from $\mathcal{B}$ to $\mathcal{B}'$ as

$$S_{\mathcal{B} \rightarrow \mathcal{B}'} = [\vec{w}_1 \cdots \vec{w}_n]^{-1} [\vec{v}_1 \cdots \vec{v}_n]$$

Lemma 7.6 letting $\mathcal{B}$ and $\mathcal{B}'$ be as above and letting $(S_{\mathcal{B} \rightarrow \mathcal{B}'})_{i,j} = \Delta_{ij}$. Then for each $1 \leq j \leq n$, $\vec{v}_j = \sum_{i=1}^n \Delta_{ij} \vec{w}_i$.

Proof: As $S_{\mathcal{B} \rightarrow \mathcal{B}'} = [\vec{w}_1 \cdots \vec{w}_n]^{-1} [\vec{v}_1 \cdots \vec{v}_n]$

$$[\vec{v}_1 \cdots \vec{v}_n] = [\vec{w}_1 \cdots \vec{w}_n] \begin{bmatrix} \Delta_{11} & \cdots & \Delta_{1n} \\ \vdots & \ddots & \vdots \\ \Delta_{n1} & \cdots & \Delta_{nn} \end{bmatrix}$$

and taking the j th column of both sides

$$\vec{v}_j = \sum_{i=1}^n \Delta_{ij} \vec{w}_i \quad \text{as desired.}$$

Lemma 7.7 letting $\mathcal{B}$ and $\mathcal{B}'$ be as above
and letting $\vec{x} \in V$ we have that

$$[\vec{x}]_{\mathcal{B}'} = S_{\mathcal{B} \rightarrow \mathcal{B}'} [\vec{x}]_{\mathcal{B}}$$

Proof: Set $[\vec{x}]_{\mathcal{B}} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$ and $[\vec{x}]_{\mathcal{B}'} = \begin{bmatrix} x_1' \\ \vdots \\ x_n' \end{bmatrix}$

so that $\vec{x} = \sum_{i=1}^n x_i \vec{v}_i$ and $\vec{x} = \sum_{i=1}^n x_i' \vec{w}_i$

Now as $\vec{x} = \sum_{j=1}^n x_j \vec{v}_j = \sum_j x_j \left(\sum_i a_{ij} \vec{w}_i \right)$

$$= \sum_j \sum_i x_j a_{ij} \vec{w}_i$$

$$= \sum_i \left(\sum_j \underbrace{a_{ij} x_j}_{x_i'} \right) \vec{w}_i$$

we have that $x_i' = \sum_j a_{ij} x_j$ for all i

and thus $\begin{bmatrix} x_1' \\ \vdots \\ x_n' \end{bmatrix} = \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$

or $[\vec{x}]_{\mathcal{B}'} = S_{\mathcal{B} \rightarrow \mathcal{B}'} [\vec{x}]_{\mathcal{B}}$

Theorem 7.8

$$i. \quad S_{\mathcal{B}' \rightarrow \mathcal{B}} = (S_{\mathcal{B} \rightarrow \mathcal{B}'})^{-1}$$

$$ii. \quad S_{\mathcal{B}' \rightarrow \mathcal{B}''} S_{\mathcal{B} \rightarrow \mathcal{B}'} = S_{\mathcal{B} \rightarrow \mathcal{B}''}$$

Theorem 7.9 (Change of Basis formula)

let $\mathcal{B}$ and $\mathcal{B}'$ be two bases for V and let $T \in \mathcal{L}(V, V)$. Then

$$\begin{aligned} [T]_{\mathcal{B}'} &= S_{\mathcal{B} \rightarrow \mathcal{B}'} [T]_{\mathcal{B}} S_{\mathcal{B}' \rightarrow \mathcal{B}} \\ &= S_{\mathcal{B} \rightarrow \mathcal{B}'} [T]_{\mathcal{B}} (S_{\mathcal{B} \rightarrow \mathcal{B}'})^{-1} \end{aligned}$$

Diagram illustrating the change of basis formula for the matrix representation of a linear transformation T . The diagram shows the relationship between the coordinate vectors of $\vec{x}$ in bases $\mathcal{B}'$ and $\mathcal{B}$, and the coordinate vectors of $T\vec{x}$ in the same bases. The blue arrows represent the change of basis matrices $S_{\mathcal{B}' \rightarrow \mathcal{B}}$ and $S_{\mathcal{B} \rightarrow \mathcal{B}'}$, and the matrix representation $[T]_{\mathcal{B}}$. The green arrow represents the matrix representation $[T]_{\mathcal{B}'}$.

Corollary 7.10 let $\mathcal{B} = \{\vec{v}_1, \dots, \vec{v}_n\}$ be a basis for $\mathbb{R}^n$ and let $T \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^n)$. Then

$$[T]_{\mathcal{B}} = \underbrace{[\vec{v}_1 \dots \vec{v}_n]^{-1}}_{S_{\{\vec{e}_i\} \rightarrow \mathcal{B}}} [T]_{\{\vec{e}_i\}} [\vec{v}_1 \dots \vec{v}_n]$$

Exercises:

2) Let $T \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^2)$ be defined by

$$T\vec{e}_1 = 3\vec{e}_1 + 5\vec{e}_2, \quad T\vec{e}_2 = 2\vec{e}_1 - \vec{e}_2$$

Now letting $\vec{v}_1 = \begin{bmatrix} 4 \\ 1 \end{bmatrix}$ and $\vec{v}_2 = \begin{bmatrix} -5 \\ -1 \end{bmatrix}$

be a new basis for $\mathbb{R}^2$, compute

$[T]_{\{\vec{v}_1, \vec{v}_2\}}$ and evaluate $[T(3\vec{v}_1 + 2\vec{v}_2)]_{\{\vec{v}_1, \vec{v}_2\}}$

(i.e. express $T(3\vec{v}_1 + 2\vec{v}_2)$
as a linear combination
of $\vec{v}_1$ and $\vec{v}_2$)

As $[T]_{\{\vec{e}_i\}} = \overset{T\vec{e}_1 \quad T\vec{e}_2}{\begin{bmatrix} 3 & 2 \\ 5 & -1 \end{bmatrix}}$ we use Cor 7.10

to obtain $[T]_{\mathcal{B}} = \begin{bmatrix} 4 & -5 \\ 1 & -1 \end{bmatrix}^{-1} \begin{bmatrix} 3 & 2 \\ 5 & -1 \end{bmatrix} \begin{bmatrix} 4 & -5 \\ 1 & -1 \end{bmatrix}$

$$= \begin{bmatrix} -1 & 5 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ 5 & -1 \end{bmatrix} \begin{bmatrix} 4 & -5 \\ 1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 5 \\ -1 & 4 \end{bmatrix} \begin{bmatrix} 14 & -17 \\ 19 & -24 \end{bmatrix}$$

$$= \begin{bmatrix} 81 & -103 \\ 62 & -79 \end{bmatrix}$$

$$\text{Now } [T(3\vec{v}_1 + 2\vec{v}_2)]_{\{\vec{v}_i\}} = \begin{bmatrix} 81 & -103 \\ 62 & -79 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \end{bmatrix}_{\{\vec{v}_i\}}$$

$$T(3\vec{v}_1 + 2\vec{v}_2) = 37\vec{v}_1 + 28\vec{v}_2 = \begin{bmatrix} 37 \\ 28 \end{bmatrix}_{\{\vec{v}_i\}}$$

$$= 37\vec{v}_1 + 28\vec{v}_2$$

3) let $S = \{f \in \mathbb{P}_3(\mathbb{R}) : \int_0^1 f = 0\}$ and

let $\mathcal{B} = \{x - \frac{1}{2}, x^2 - \frac{1}{3}, x^3 - \frac{1}{4}\}$ and

$\mathcal{B}' = \{x - \frac{1}{2}, x^2 - \frac{2x}{3}, x^3 - \frac{3x^2}{4}\}$. Find

the change of basis matrix.

We first represent the elements in $\mathcal{B}$ as linear combinations of the elements $\mathcal{B}'$

i. $x - \frac{1}{2} = 1(x - \frac{1}{2})$ $\Delta_{11} = 1, \Delta_{21} = 0, \Delta_{31} = 0$

ii. $x^2 - \frac{1}{3} = 1(x^2 - \frac{2x}{3}) + \frac{2}{3}(x - \frac{1}{2})$ $\Delta_{22} = 1, \Delta_{12} = \frac{2}{3}$

iii. $x^3 - \frac{1}{4} = 1(x^3 - \frac{3x^2}{4}) + \frac{3}{4}(x^2 - \frac{2x}{3}) + \frac{1}{2}(x - \frac{1}{2})$ $\Delta_{33} = 1, \Delta_{23} = \frac{3}{4}, \Delta_{13} = \frac{1}{2}$

Thus $S_{\mathcal{B} \rightarrow \mathcal{B}'} = \begin{bmatrix} 1 & 2/3 & 1/2 \\ 0 & 1 & 3/4 \\ 0 & 0 & 1 \end{bmatrix}$

Projections (chapter 4 of Strong or 6.C in Axler)

Def 7.11 A linear map $T \in \mathcal{L}(V, V)$ is a projection iff $T(T(\vec{x})) = T(\vec{x})$ for $\vec{x} \in V$. Such a map satisfying $T \circ T = T^2 = T$ is called idempotent.

Exercise:

4) Show that $T \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^2)$ defined by

$$T\vec{e}_1 = -3\vec{e}_1 - 6\vec{e}_2, \quad T\vec{e}_2 = 2\vec{e}_1 + 4\vec{e}_2$$

is a projection.

$$\begin{aligned} \text{Since } T(T\vec{e}_1) &= T(-3\vec{e}_1 - 6\vec{e}_2) \\ &= -3T\vec{e}_1 - 6T\vec{e}_2 \\ &= 9\vec{e}_1 + 18\vec{e}_2 - 12\vec{e}_1 - 24\vec{e}_2 \\ &= -3\vec{e}_1 - 6\vec{e}_2 = T\vec{e}_1 \end{aligned}$$

and similarly $T(T\vec{e}_2) = T\vec{e}_2$ we have that $T^2 = T$ and thus T is a projection.

Def 7.12 A matrix $P \in F^{n \times n}$ is a projection

matrix iff $p^2 = p$.

Examples: $p_x = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$, $p_y = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$, $[T] = \begin{bmatrix} -3 & 2 \\ -6 & 4 \end{bmatrix}$